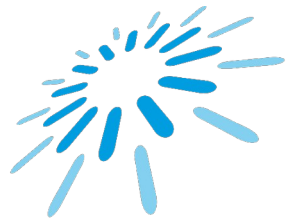




Software Architecture and Design Patterns

Prof. Dr. Jóakim von Kistowski

Lecture 02



TH Aschaffenburg
university of applied sciences

	Monday	Tuesday	Wednesday	Thursday	Friday
08:00 - 09:30					Vorlesung Software Design
09:45 - 11:15					Übung Software Design
11:30 - 13:00			Lecture Software Design International		
	Lunch Break				
14:00 - 15:30			Exercises Software Design International		
15:45 - 17:15					
17:30 - 19:00					

**What is the difference between Micro-Architecture
and Macro-Architecture?**

What are the major skills a Software Architect should have?

Software Architecture: Quality



Recap: Quality and ISO 25010

Ranking Quality by Example

Quality Scenarios

What is Software Quality?

What is Quality? (1/4)

- Degree to which the system satisfies the **stated** and **implied** needs of its **various stakeholders** [ISO 25010].
- Degree to which a set of inherent characteristics fulfills requirements [after ISO 9000:2000]

What is Quality? (2/4)

- Degree to which the system satisfies the **stated** and **implied** needs of its **various stakeholders** [ISO 25010].
- Degree to which a set of inherent characteristics fulfills requirements [after ISO 9000:2000]
- You want to buy a car:
 - Which requirements would you mention explicitly?
 - Which requirements would you take for granted?

What is Quality? (3/4)

- Degree to which the system satisfies the **stated** and **implied** needs of its **various stakeholders** [ISO 25010].
- Degree to which a set of inherent characteristics fulfills requirements [after ISO 9000:2000]
- You want to buy a car:
 - Which requirements would you mention explicitly?
 - Which requirements would you take for granted?
- Example Chat App:
 - Which requirements would you mention explicitly?
 - Which requirements would you take for granted?

What is Quality? (4/4)

- Degree to which the system satisfies the **stated** and **implied** needs of its **various stakeholders** [ISO 25010].
- Degree to which a set of inherent characteristics fulfills requirements [after ISO 9000:2000]
- Aspects of quality relate to requirements
 - Functional requirements
(e.g., Domain specifics, Correctness, Suitability)
 - Non-functional requirements
(e.g., quality requirements, performance, IT-security, usability, ...)

ISO 25010

The standard on software quality



Software Architecture: Quality



Recap: Quality and ISO 25010

Ranking Quality by Example

Quality Scenarios

Quality according to ISO 25010

Rank the Top-3 Quality Attributes for our Example Systems (+ at least one unimportant attribute)

TeaStore



Online-Shooter



Try to use as many of the sub-categories as possible

Write your answers down. You will need them later.



ITSQB Glossary

Question

How can we check if a software architecture is good/bad for a specific quality attribute?

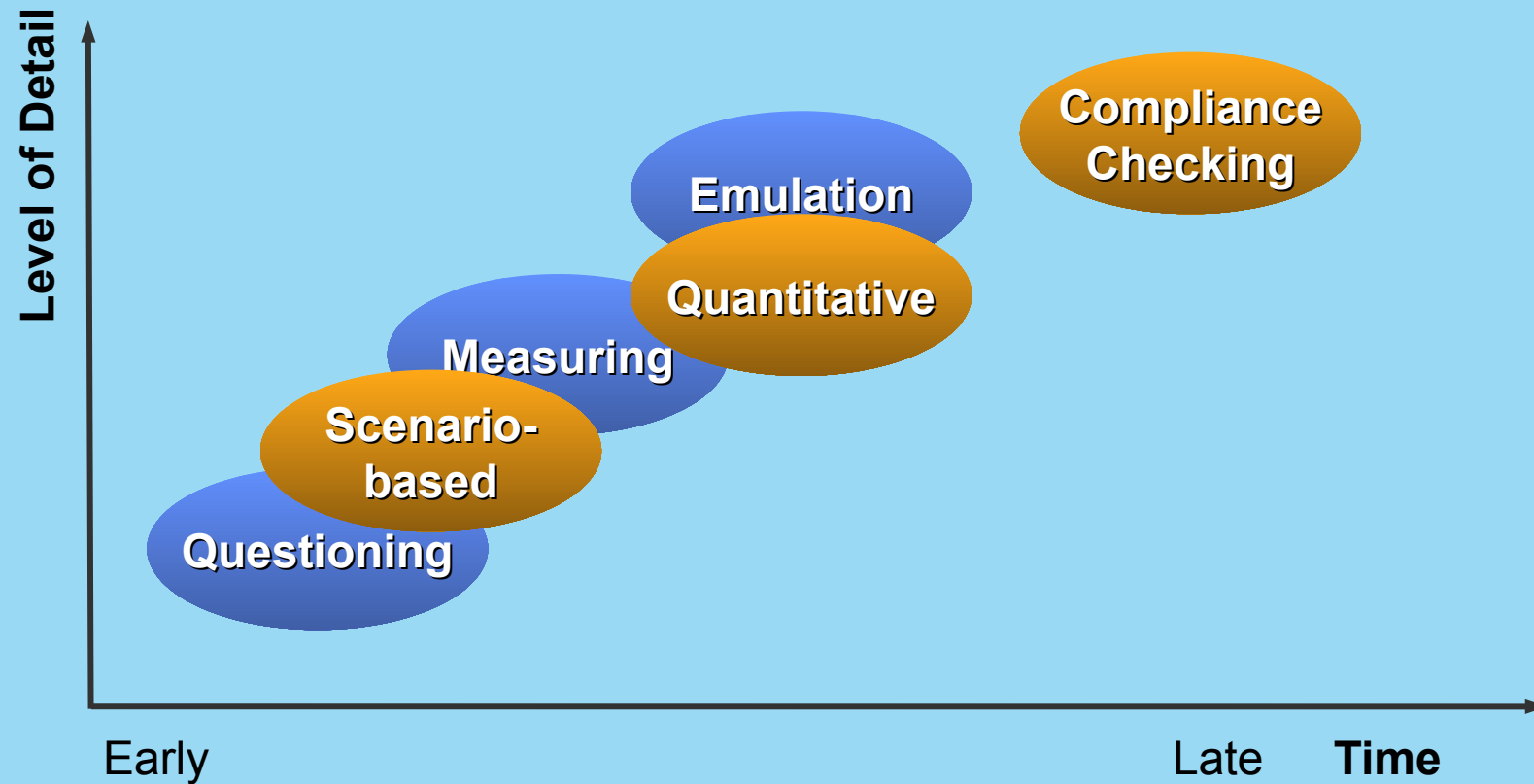
Question of the day:

“How do we specify measurable quality goals for our architecture?”

Put another way:

„How do we check if a software architecture is good?“

Architecture Evaluation Techniques



Software Architecture: Quality



Recap: Quality and ISO 25010

Ranking Quality by Example

Quality Scenarios

Quality Attribute Scenarios

Scenario:

„A short statement describing an interaction of one of the stakeholders with the system“
(Clements et. al)

Examples:

- User: Uses the system to perform a certain task (Use-Case)
- Maintainer: Change to the system (e.g.: OS-Upgrade)
- Developer: Uses the architecture, to build systems; predicts Quality of Service
- Product-line-manager: Uses a reference architecture to create a second product for a given product line

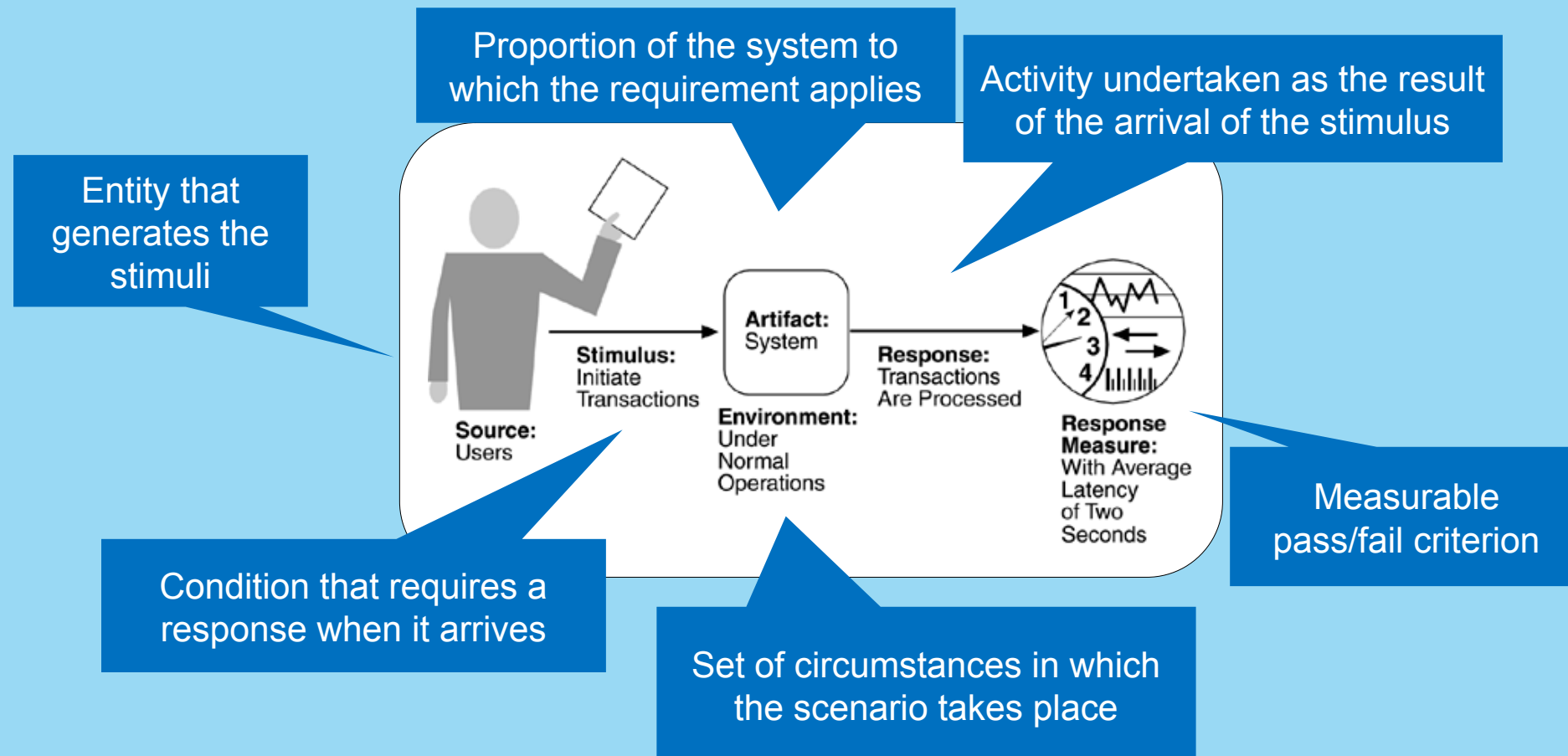
Scenarios and prioritization are defined by the stakeholders.

Scenario-based Architecture Evaluation

Architecture can be evaluated by analyzing the effects of a scenario and the architecture's „response“.

- How to implement the scenario with the given architecture?
- Can all (quality-)requirements be met? If no: why?
- What changes have to be made in order to meet the szenario's requirements?
- Which design decisions have to be made?
- How much work is it? How many components have to be changed?
- What risks result from meeting the scenaro's requirements?

Quality Attribute Scenario Notation



Quality Attribute Scenario Notation (again, slightly different)

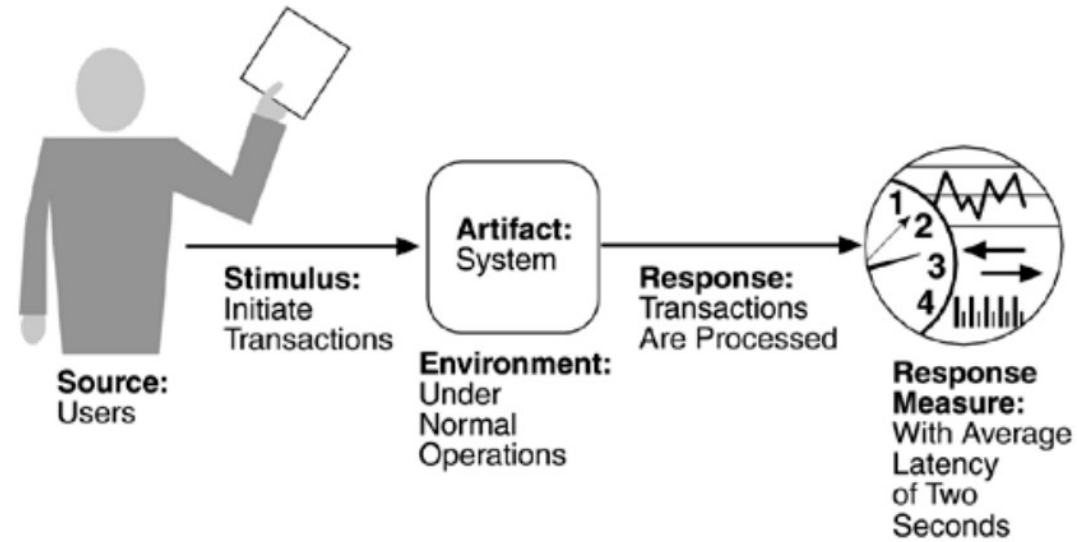
What is a scenario?



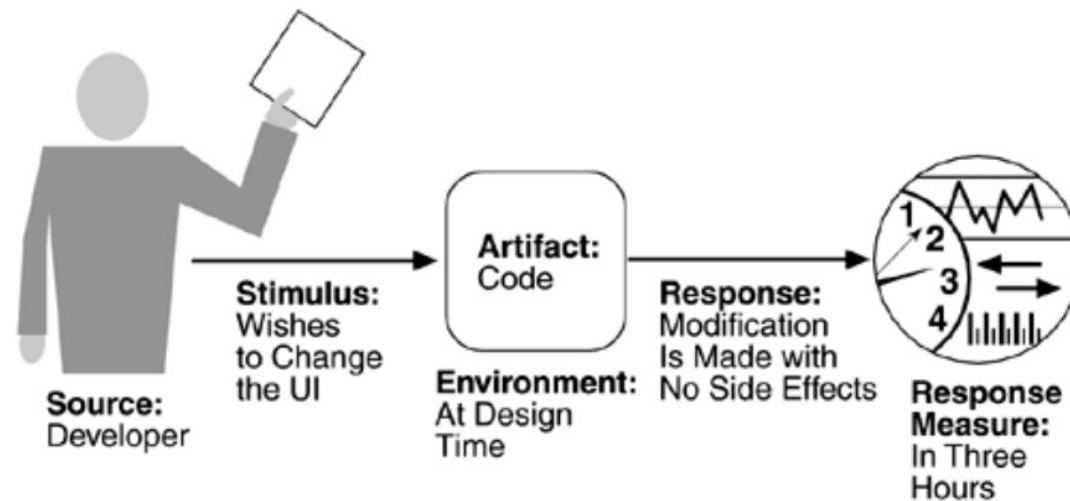
- The **Artifact** is at the center, representing the product. Artifacts can be the „System“ as well as concrete building blocks, subsystems, or Microservices.
- The **Source** denotes the trigger for the given action – e.g., a user, a persona or an external system. The source sets the context for the stimulus regarding the artifact.
- The **Stimulus** describes the action on the product. This can be a use-case or function call.
- The **Response** describes the artifact's answer or reaction to the stimulus.
- The **Response Measure** formulates an evaluation on how well the scenario has been completed – e.g., a target time or another metric.
- The **Environment** is used to describe the context in which the artifact produces the response with it's response measure. Use this construct to describe boundaries within which certain product behavior should be expected.

Quality Attribute Scenarios: Examples (1/2)

Performance



Modifiability



Quality Attribute Scenarios: Examples (2/2)

Resiliency

(Multiplayer-Shooter)

Source	Stimulus	Artifact	Environment	Response	Measure
Server	Down (no power)	Game- Server	Before noon (few players online)	Game is resumed on a new server	2 minutes

Modifiability

(Insurance)

Source	Stimulus	Artifact	Environment	Response	Measure
Domain- Department	Finalizes prices for new insurance policies	Customer- System	No regulatory changes at the moment	New customers can by insurance with the newest policies	1 Sprint



Task – Quality Scenarios

Specify Quality Scenarios for each of our example applications.

Specify at least one scenario per top 3 quality attribute.



TeaStore



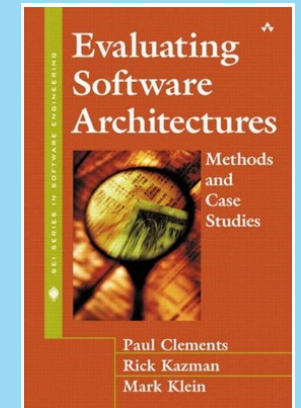
Online-Shooter



How do we get quality scenarios from our stakeholders?

Architectural Trade-Off Analysis Method (ATAM)

- Human-centric process for identifying risks early on in software design
- Focuses specifically on four quality attributes (NFPs)
 - Modifiability
 - Security
 - Performance
 - Reliability
- Purpose:
 - Reveals how well an architecture satisfies quality goals and how those goals trade-off
 - „Assess consequences of architectural decision alternatives in light of quality attribute requirements.“



ATAM – Business Drivers

- The system's critical functionality
- Any technical, managerial, economic, or political constraints
- The project's business goals and context
- The major stakeholders
- The principal quality attribute (NFP) goals

ATAM – Architecture

- Technical constraints
 - Required hardware platforms, OS, middleware, programming languages, and OTS functionality
- Any other systems with which the system must interact
- ***Architectural approaches*** that have been used to meet the quality requirements
 - Sets of architectural design decisions employed to solve a problem
 - Typically architectural patterns and styles

ATAM – Analysis

- Key step in ATAM
- Objective is to establish relationship between architectural approaches and quality attributes
- For each architectural approach a set of analysis questions are formulated
 - Targeted at the approach and quality attributes in question
- System architects and ATAM evaluation team work together to answer these questions and identify
 - Risks → these are distilled into risk themes
 - Non-Risks
 - Sensitivity points
 - Trade-off points
- Based on answers, further analysis may be performed

ATAM – Outputs

- Concise presentation of the architecture
- Articulation of the business goals
- Prioritized quality attribute requirements expressed as quality attribute scenarios
- A set of risks and nonrisks (and risk themes)
- Mapping of architectural decisions to quality requirements
- A set of identified sensitivity and tradeoff points

ATAM – Pros and Cons



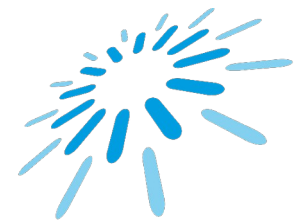
Easy to apply – for all types of stakeholders
No formal methods needed
Structured and guided approach
Explicit goals & scenarios

No formal quantification
Depends on skills of stakeholders
Requires experience & good estimation skills



Until next time!

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